

Overview of the 2020 Hyundai Santa Fe Model

The 2020 Hyundai Santa Fe model is a highly automated vehicle with several onboard technologies and services that utilise information collected, gathered, recorded or stored by the vehicle. One of the technologies incorporated into the Hyundai Santa Fe is the BlueLink telematics system. Blue Link is a dynamic, telematics technology that allows Hyundai vehicles to communicate (send and receive important and useful information). The system uses an enhanced cellular network, with automatic roaming, that optimises connections and prioritises emergency requests. It is also designed to use voice-response technology with the addition of GPS and a live assistance team for selected needs. Blue Link, amongst other things, provides:

- Automatic emergency assistance, in the unfortunate event of a collision. This includes forward collision warning with pedestrian detection and automatic braking.
- The convenience of Destination Search by Voice, as well as the ability to remotely operate various vehicle features.
- On demand diagnostics and vehicle analytics feedback features which are essentially activated online.
- Notification features and feedback mechanism (activatable online using configuration enabling suites).
- The vehicle is equipped with an event data recorder (EDR) which serves the purpose of recording crash footage or near crash-like situations such as in the case of air bag deployment or hitting a road obstacle, data that will assist in understanding how a vehicle's system performed.

The 2020 version of the Hyundai Santa Fe comes with smart sense suite of active safety components and features: a 7.0-inch touchscreen infotainment system and navigation are all available with the vehicle with Android Auto and Apple CarPlay capability, four USB ports, and a backup camera. The Hyundai Smart Sense safety features as standard equipment. It has a satellite radio, LED lighting, wireless device charging and an infinity stereo system. Other notable components include adaptive cruise control, a driver inattention monitor, and lane keeping assistance.

At the centre of its technological novelty, the 2020 Hyundai Santa Fe model relies on a number of crucial connected and autonomous elements. Some of these components that form part of the case study are highlighted below.

BlueLink Telematics System

The BlueLink Telematics System is a key component incorporated into the Hyundai Santa Fe that gives drivers greater control over vehicle functions and features. This is designed to meet 3 distinct operational package requirements including but not limited to: connected care (e.g., safety and car care), Remote services (e.g., Remote access control, recovery) and Guidance/convenient features (e.g., search by voice). The system comes with a personalised Google-based point-of-interest search navigation system which makes destination and location finding easier. Also incorporated is Bluetooth connectivity, Apple CarPlay and Android Auto capabilities. The system is able to keep a catalogue of destinations and frequency of visits while highlighting regularly visited areas for fully automated anticipation for future journeys along the same route. For instance, should you be heading to the University Campus at a regular time each day, the software is able to log your trips and even offer traffic advice in relation to departure time. This feature operates using third party apps accessible through MyHundai.com, the My Hyundai App or Google store. There are also a number of native apps for use with the vehicle that enhance control of several features from personal smartphones. There is

also the Destination Search by voice button feature to search for most up to date locations. In addition, the Apple's Siri digital assistant is accessible by the telematics system as well for seamless answering of calls, replying to texts and even browsing the internet on the go – all handsfree. This provides not only a safer driving experience but also sustained updates and interconnectedness accessible via Apple Car Play.

***Reference manuals/ sections:**

- **The Hyundai Santa Fe Quick Reference Guide:** BlueLink Telematics System (page 22,) Android auto (page 28), Navigation (page 25), Third party App (page 32), Tire pressure monitoring system (page 40).
- **BlueLink User's Manual Display Audio Navigation System:** Smart phone Mobile Application (pages 9 - 10), In-vehicle Wi-Fi (page 13), BlueLink Connected Care (safety & Car Care) (pages 14 - 21), Blue Link Remote Package (accessible via mobile app) (pages 22 - 26), Controls for Blue Link in vehicle voice-response (page 11), Automatic Diagnostic Trouble Code Notification (pages 18 - 20), Remote Door Unlock/Lock (page 22), Remote start and climate control (pages 24 - 25).
- **Hyundai Owner's Manual – Operations Maintenance and specifications:** Vehicle Data Collection and Event Data Recorders – (page F10), Audio systems/ Navigation system (page 4-4), Automatic Climate Control System (page 3-146), wireless cellular phone charging system (page 3-174), smart cruise controls (page 5-106), smart cruise control vehicle to vehicle distance (page 5-111).

Blind Spot Detection System with “Safe-Exit” Feature

The Blind Spot Detection (BSD) or Blind- Spot Collision Warning (BCW) in the Hyundai Santa Fe measures the speed of and distance from vehicles approaching from the rear by using two magnetic wave radar sensors attached in the rear bumper. The BSD/BCW detects vehicle within the blind spot zone and gives off an alarm (visual and audible). The two magnetic radar sensors provide these functions for BSD/BCW by sensing other vehicles in the BSD/BCW zone and turning on the BSD/BCW warning lamp for the driver. The warning lamp in synchrony starts blinking, along with an audible sound when the driver turns on the turn signal lamp to enter the lane where another vehicle is driving. Blind Spot Monitoring systems employ radar modules located at each rear quarter panel of the bumper or cameras (depending on the trim) designed to scan for and detect vehicles in adjacent lanes. The Santa Fe has warning lights in the exterior mirrors that illuminate or flash when a vehicle is detected close by and one lane over. Blind spot monitors as a safe-exit feature generally work well and can help prevent the vehicle from cutting off another driver. The Blind spot monitoring with rear cross-traffic alert is no longer standard on base models of the Santa Fe, but is available on higher trim levels including the 2020 model which has a Blind-View monitor. The Blind-View monitor displays a camera feed of the vehicle's blind spot in the digital gauge cluster when the corresponding turn-signal is activated.

***Reference manuals/ sections.**

- **Hyundai Owner's Manual – Operations Maintenance and specifications:** Blind spot Collision Warning (BCW)/Blind spot Collision Avoidance Assist switch (BCA) (pages 5-68 to 5-77).
- **The Hyundai Santa Fe Quick Reference Guide:** Safe Exit Assist (SEA) (page 35).

Rear-view Camera with Cross-Traffic Detection, Rear Collision Avoidance and Smart Trunk Hatch

The Hyundai Santa Fe employs a similar mechanism to that used for blind spot monitoring for rear cross-traffic alert to warn the driver of vehicles or pedestrians approaching the rear of the vehicle from the side. The Rear Cross traffic Collision Avoidance Assist (RCCA) system uses radar sensors to monitor approaching vehicles to the left and right while the vehicle is in reverse. This feature comes in particularly handy when backing out of parking spots in crowded lots. Some variants of the 2020 Hyundai Santa Fe also apply the brakes automatically to stop the vehicle in the event of a possible collision. This automated process is achieved where the instrument cluster informs the driver of the brake control. If the rear view mirror system is in activation mode, a message also appears on the audio or AVN screen.

***Reference manuals/ sections.**

- **The Hyundai Santa Fe Quick Reference Guide:** Rear Cross-Traffic Collision- Avoidance Assist (page 35).

Smart Cruise Control with Forward Collision Avoidance

The Hyundai Santa Fe has a smart cruise control that is adaptive for convenience. It operates using the vehicle's radars and camera modules to alter the set cruising speed if it detects a slower vehicle ahead. When the adaptive cruise control is engaged, the car maintains a specific distance from the car in front. Depending on the 2020 trim, the vehicle comes in three set options with the most advanced option able to bring the vehicle to a complete stop if the car ahead stops. The feature is designed to conveniently reduce driver fatigue. The Smart Cruise Control system, amongst other things, allows the driver to program the vehicle to maintain constant speed and minimum distance between the vehicle ahead. It automatically adjusts the vehicle speed to maintain the programmed speed and following distance without requiring the driver to depress the accelerator or brake pedals. Using forward facing cameras and radar, the vehicle's automated emergency braking kicks in to warn the driver of an imminent forward collision with another vehicle, pedestrian or object and then brake the vehicle on behalf of the driver if he/she takes no action. This automated emergency braking system is also applicable when the vehicle is reversing. Similar to adaptive/smart cruise control, the forward-collision avoidance warning system uses radars, cameras, or both to analyse the road ahead for slow or stopped traffic. Forward-collision warning is a more basic form of collision mitigation and will alert the driver – usually with both a visual and an audible warning, that a collision is imminent, in the hope that the driver will hit the brakes.

***Reference manuals/ sections.**

- **Hyundai Owner's Manual – Operations Maintenance and specifications:** Forward Collision Avoidance (pages 5-54 to 5-67), Smart Cruise Control (pages 5-106 to 5-124).
- **The Hyundai Santa Fe Quick Reference Guide:** Smart Cruise Control/ cruise control (Page 41), Forward Collision – Avoidance Assist (page 34).

Lane Assist System with Driver Attention Warning

The lane Assist System comprises of both lane departure warning and lane keeping assist. In the former, a camera-based driver assistance feature looks for lane markings and alerts the driver if the vehicle is in danger of wandering out of its lane or has begun to exit the lane without the turn signal being activated. The Lane keeping assist, unlike the lane departure warning, helps prevent lane departure by making small corrections to the steering to keep the vehicle in the intended lane. While some systems still provide a visual or an audible alert to accompany the lane departure, other variants of the Hyundai will just quietly recentre the vehicle in the lane. The Driver Attention Warning system keeps an eye on the driver to help prevent accidents caused by drowsy and distracted driving. It operates using sensors to monitor the car's movements and volume of steering corrections to determine if the driver is attentively driving or is tired or texting. The Driver Attention Warning system provides a visual or audible alert to encourage taking a break from driving to perk up. More modern variants of the 2020 model have driver-facing cameras that watch for signs of sleepiness.

***Reference manuals/ sections.**

- **Hyundai Owner's Manual – Operations Maintenance and specifications:** Lane keeping assist system and Drive Attention Warning (pages 5-93 to 5-105).
- **The Hyundai Santa Fe Quick Reference Guide:** Lane Keeping Assist (page 33).